

## **Chapter 6 Part III – Absorption and Variable Costing**

# Variable costing and Absorption Costing



|                                                |
|------------------------------------------------|
| Describe absorption costing                    |
| Describe variable costing                      |
| What is a contribution margin income statement |
| What is an absorption costing income statement |
|                                                |
|                                                |
|                                                |

## Absorption Costing

---

All manufacturing-related costs (fixed and variable) are absorbed into the cost of the product.

---

DM, DL, and MOH are treated as product costs.

---

Required by GAAP and the IRS.

# Variable Costing

Only variable manufacturing costs are treated as product costs.

Only used for *internal* management purposes.

Often leads to better decisions:

Shows incremental cost of manufacturing each unit.

Operating income will not be affected by changes in inventory levels.

# Example Absorption vs. Variable Costing

---

## Variable costs:

|                                                 |      |
|-------------------------------------------------|------|
| Direct material cost per unit produced .....    | \$35 |
| Direct labor cost per unit produced .....       | \$10 |
| Variable MOH cost per unit produced .....       | \$5  |
| Variable operating expenses per unit sold ..... | \$2  |

## Fixed costs:

|                                |             |
|--------------------------------|-------------|
| Fixed MOH .....                | \$1,000,000 |
| Fixed operating expenses ..... | \$300,000   |

## Other information:

|                            |              |
|----------------------------|--------------|
| Units produced .....       | 40,000 units |
| Sales price per unit ..... | \$100        |

---

## Example Absorption vs. Variable Costing – differences in presenting fixed manufacturing overheads

|   | A                                                    | B                  | C                |
|---|------------------------------------------------------|--------------------|------------------|
| 1 | Manufacturing Costs Per Unit                         | Absorption Costing | Variable Costing |
| 2 | Direct material cost per unit                        | \$ 35              | \$ 35            |
| 3 | Direct labor cost per unit                           | 10                 | 10               |
| 4 | Variable MOH cost per unit                           | 5                  | 5                |
| 5 | Fixed MOH cost per unit (\$1,000,000 ÷ 40,000 units) | 25                 | 0                |
| 6 | Total cost per unit                                  | \$ 75              | \$ 50            |
| 7 |                                                      |                    |                  |

Under variable costing, no fixed MOH is assigned to the product cost.

|   | A                                                                     | B                  | C                |
|---|-----------------------------------------------------------------------|--------------------|------------------|
| 1 | Operating Expenses of the Period                                      | Absorption Costing | Variable Costing |
| 2 | Variable operating expenses when 40,000 units are sold (40,000 × \$2) | \$ 80,000          | \$ 80,000        |
| 3 | Fixed operating expenses                                              | 300,000            | 300,000          |
| 4 | Fixed MOH                                                             | 0                  | 1,000,000        |
| 5 | Total operating expenses (period costs)                               | \$ 380,000         | \$ 1,380,000     |
| 6 |                                                                       |                    |                  |

Under variable costing, fixed MOH is treated as a period cost.

## Example Absorption vs. Variable Costing – difference in the decisions

---

Assume the company decides to produce an extra 5,000 units.

---

Variable costing (5,000 units x \$50) = \$250,000.

---

Absorption costing (5,000 units x \$75) = \$375,000.

---

Variable costing is more accurate for internal decision making.

# The Traditional Income Statement versus Contribution Margin Income Statement

| Traditional Income Statement (amounts in \$) |                  |
|----------------------------------------------|------------------|
| Sales (40,000*\$100)                         | 4,000,000        |
| Less:                                        |                  |
| Cost of goods sold (40,000*\$75)             | 3,000,000        |
| <b>Gross Profit</b>                          | <b>1,000,000</b> |
| Less:                                        |                  |
| Operating expenses                           | 380,000          |
| <b>Operating Income</b>                      | <b>620,000</b>   |

| Contribution Margin Income Statement (amounts in \$) |                  |
|------------------------------------------------------|------------------|
| Sales (40,000*\$100)                                 | 4,000,000        |
| Less variable expenses                               |                  |
| Variable Cost of goods sold (40,000*\$50)            | 2,000,000        |
| Variable operating expenses                          | 80,000           |
| <b>Contribution Margin</b>                           | <b>1,920,000</b> |
| Less fixed expenses                                  |                  |
| Fixed Mfg OH                                         | 1,000,000        |
| Fixed operating expenses                             | 300,000          |
| <b>Operating income</b>                              | <b>620,000</b>   |



# The Contribution Margin Income Statement - Characteristics

---

Organized by cost behavior.

---

All variable costs are expensed **above** the contribution margin line.

---

All fixed costs are expensed **below** the contribution margin line.

---

**Contribution margin = sales revenue – variable expenses.**

---

Operating income is the same in both statements when all units are produced and sold in same period.

---

Only for internal use.

# **The Contribution Margin Income Statement**